# Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

## Summary

Software engineer with 2+ years building **ML/AI systems**, **computer vision pipelines**, and **production applications** in Python and C++. Experience implementing research techniques with transformers, optimizing performance, and deploying scalable systems.

## Education

**University of Southern California, Los Angeles, CA** **August 2024 – May 2026**
*Master of Science in Computer Science* *GPA: 3.75/4.0*
Coursework: Analysis of Algorithms, Advanced NLP, Computer Vision, Machine Learning

**Pandit Deendayal Energy University, Gandhinagar, India** **November 2020 – May 2024**
*Bachelor of Technology in Computer Science and Engineering* *GPA: 3.88/4.0*
Coursework: Computer Graphics, Image Processing, Deep Learning, Algorithms

## Experience

**USC Dornsife** **Los Angeles, CA**
*Software Engineer* *September 2025 – Present*

- Built **production backend systems in Python** with async processing pipelines, implementing optimizations that reduced latency from **1.2s to 350ms** and increased throughput by 3x under concurrent workloads.
- Developed **scalable data processing infrastructure** handling 5K+ monthly jobs with performance tuning, achieving 99%+ reliability while collaborating with cross-functional teams on end-to-end solutions.

**Grad8** **Los Angeles, CA**
*Software Engineer Intern* *June 2025 – August 2025*

- Implemented **computer vision pipelines integrating multiple ML models** (GPT-4o Vision, Claude Vision, OCR) for image processing, optimizing inference to handle 10K+ daily requests with efficient resource utilization.
- Designed **concurrent batch-processing engine** using multithreading, processing 1K+ files per batch with fault-tolerant execution and clear technical documentation for research-to-production workflows.

**USC AD Labs** **Los Angeles, CA**
*ML Research Assistant* *January 2025 – May 2025*

- Built a **GPT-based transformer model for autonomous systems** using PyTorch, implementing research techniques from papers to improve trajectory planning by 18% with end-to-end training pipelines on 100K+ records.
- Architected a **spatial-temporal encoder** integrating geometric data with agent context via multi-head attention, demonstrating ability to implement complex algorithms from research and optimize for real-world deployment.

**Aasma Technology Solutions** **India**
*Software Developer Engineer - (SecureSync)* *August 2023 – July 2024*

- Engineered **performance-optimized APIs and data pipelines** using Python, improving response times by 30% and pipeline efficiency by **68%** through algorithmic optimization and resource-efficient design.
- Built **cross-platform software with automated CI/CD** using Jenkins and GitHub, delivering production-quality code with comprehensive testing in Agile sprints collaborating with engineering teams.

**Aasma Technology Solutions** **India**
*Full Stack Developer Intern - (VibrantChem)* *May 2023 – August 2023*

- Developed **real-time streaming systems** using WebSockets for 2M+ records, implementing efficient data synchronization and compression techniques to minimize bandwidth and memory requirements.
- Built **scalable microservice architecture** with Docker deployments, collaborating with cloud and content teams to design end-to-end solutions that scaled to production workloads.

## Skills

**Languages**: Python, C++, JavaScript/TypeScript, SQL, CUDA basics
**ML/CV**: PyTorch, Transformers, CLIP, Stable Diffusion, computer vision, neural networks, embeddings
**Infrastructure**: Docker, AWS, GCP, Kubernetes, CI/CD, distributed systems, performance optimization
**Practices**: Research implementation, algorithm optimization, Agile/Scrum, technical documentation

## Projects

**PiXMod** Live | *CLIP, Stable Diffusion, SAM, PyTorch, MongoDB*

- Built an **AI-powered image processing platform** using transformer models (CLIP, Stable Diffusion, SAM) for segmentation, generation, and semantic search across 100K+ embeddings with optimized inference.

**CaptionCraft** Live | *Transformers, PyTorch, BLIP, ViT-GPT-2, Flask*

- Benchmarked **multiple transformer architectures** (BLIP, GIT, ViT-GPT-2) on 10K images/100K captions, implementing research techniques to improve caption accuracy from 81% to 96% with scalable inference APIs.

**Big-Brain** Live | *LLaMA-2, FAISS, Pinecone, GCP*

- Built a **semantic retrieval system** using transformer embeddings and vector search (FAISS), processing 50K+ documents with optimized similarity search pipelines achieving sub-second retrieval latency.